

Muhammad Akbar Khan



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RESEARCH PROFILE

MS Applied Mathematics researcher at NED University specializing in physics-informed neural networks and data-driven PDE solvers. First author or co-author of five research papers: one published in *Machine Learning: Science and Technology* (IOP Publishing) on an RFF–eikonal joint constraint framework for level-set interface advection; one under peer review at *Scientific Reports* (Springer Nature) on a Physics-Informed Neural Operator (PINO/FNO) for thermal ranking of low-cost wall materials in hot-dry climates; two papers under review at *Machine Learning: Science and Technology*, one on SDF-Aware Weighting with adaptive eikonal regularisation for three-dimensional level-set PINNs (MLST-105968) and one on a Fourier Neural Operator surrogate for hygrothermal ranking of indigenous wall materials (MLST-106107); and one preprint on a data-free Physics-Informed Neural Operator for level-set interface advection. Seeking a fully-funded PhD position to advance scientific machine learning at the intersection of numerical analysis and deep learning.

RESEARCH INTERESTS

Physics-Informed Neural Networks (PINNs) · Neural Operators · Scientific Machine Learning · Data-Driven PDE Modeling · Numerical Analysis · Level-Set Methods for Multiphase Flow · Computational Heat Transfer

EDUCATION

NED University of Engineering & Technology

Aug 2024 – Dec 2026

Master of Science (MS) in Applied Mathematics

Karachi, Pakistan

- CGPA: **3.44/4.00**
- Thesis: *A Systematic Study of Physics-Informed Neural Networks for the Level-Set Interface Advection*
- Supervisor: Dr. Fahim Raees (Associate Professor; PhD, TU Delft, 2016)
- Coursework:
 - *Core mathematics:* Differential Equations, Linear Algebra, Advanced Discrete Mathematics
 - *Computational methods & optimization:* Numerical Methods and Applications, Scientific Computing, Operations Research and Optimization
 - *Statistics & machine learning:* Applied Statistics, Statistical Method and Data Analysis, Fuzzy Logic and Neural Networks

International Islamic University Islamabad

Jan 2017 – Jan 2021

Bachelor of Science (BS) in Mathematics

Islamabad, Pakistan

- CGPA: **3.37/4.00**
- **HEC Need-Based Scholarship** (Higher Education Commission, Pakistan): full tuition and stipend for the duration of the BS program.

PUBLICATIONS

[1] **M. Akbar Khan** and F. Raees. *A Systematic Study of Physics-Informed Neural Networks for the Level-Set Interface Advection*. **Machine Learning: Science and Technology** (IOP Publishing), 2026. [doi:10.1088/2632-2153/ae8b74](https://doi.org/10.1088/2632-2153/ae8b74). Published July 2026. Gold Open Access, CC BY 4.0.

[2] **M. Akbar Khan**, F. Raees, and U. Fatima. *A Physics-Informed Neural Operator for Thermal Ranking of Low-Cost Wall Materials in Hot-Dry Climates*. **Scientific Reports** (Springer Nature), 2026. Under peer review, July 2026.

[arXiv:2607.25668](#).

[3] **M. Akbar Khan**. *SDF-Aware Weighting: Adaptive Eikonal Regularisation for Three-Dimensional Level-Set Physics-Informed Neural Networks*. **Machine Learning: Science and Technology** (IOP Publishing), 2026. Under review; submitted August 2026 (MLST-105968).

[arXiv:2608.08322](#).

[4] F. Raees and **M. Akbar Khan**. *A Fourier Neural Operator Surrogate for Hygrothermal Ranking of Indigenous Wall Materials in Hot-Dry Climates*. **Machine Learning: Science and Technology** (IOP Publishing), 2026. Under review; submitted September 2026 (MLST-106107).

[doi:10.5281/zenodo.21862964](#).

[5] **M. Akbar Khan**. *A Data-Free Physics-Informed Neural Operator for Level-Set Interface Advection*. Preprint, 2026. [arXiv:2609.38195](#).

RESEARCH EXPERIENCE

NED University of Engineering & Technology

Research Assistant, Dept. of Mathematics

March 2026 – Present

Karachi, Pakistan

- Appointed under **Sindh Research Project 321**: mathematical modeling for heat ventilation in rural housing.
- Developing finite-difference solvers and PINN models for heat transfer simulation in indigenous wall assemblies under Dr. Fahim Raees.

TEACHING EXPERIENCE

Cadet College Ormara

Lecturer of Mathematics & OI/C FBISE Affairs

Aug 2022 – Dec 2025

Gwadar, Pakistan

- Delivered mathematics lectures and tutorials at intermediate and higher-secondary level; designed and assessed examinations.
- Managed all Federal Board of Intermediate & Secondary Education (FBISE) academic affairs: exam registrations, compliance with board regulations, and direct correspondence with FBISE.

Excelsior School System

Mathematics Teacher

May 2021 – May 2022

Islamabad, Pakistan

- Planned and delivered mathematics lessons; set and marked examinations following the national curriculum.

TECHNICAL SKILLS

Scientific ML

PyTorch, Physics-Informed Neural Networks, Fourier Neural Operators, Physics-Informed Neural Operators (PINO), Random Fourier Features, L-BFGS / Adam optimization

Programming

Python (NumPy, SciPy, Pandas, Matplotlib, Statsmodels, SQLAlchemy), SQL / PostgreSQL

Numerical Methods

Finite-difference and finite-volume schemes, classical ODE/IVP solvers, spectral methods, numerical linear algebra

Tools

Git, GitHub, VS Code, PyCharm Pro, Jupyter, L^AT_EX (MiKTeX), TikZ

HPC

GPU training on Tesla T4, PyTorch 2.10, Python 3.12

OPEN-SOURCE PROJECT

PINN Level-Set Solver

[AkbarTheAnalyst/pinn-level-set](#)

PyTorch implementation underlying published MLST paper (doi:10.1088/2632-2153/ae8b74)

- Full PINN pipeline: 8-layer MLP with Tanh activations and Xavier initialization; RFF input embedding; causal weighting.
- Four canonical benchmarks (translation, rotation, reversed vortex, Zalesak disc) with automated ablation infrastructure.

CERTIFICATIONS

- AI for Everyone, DeepLearning.AI (Coursera)
- Google Data Analytics Professional Certificate, Google (Coursera)

LANGUAGES

English: Professional working proficiency · **Urdu:** Native

REFERENCES

Dr. Fahim Raees (MS Supervisor)

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NED University of Engineering & Technology
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Dr. Ubaida Fatima

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